

verify that the functions

is analytic
تحليلية

$$z = x + iy$$

$$① f(z) = |z|^2$$

$$\therefore |z| = \sqrt{x^2 + y^2}$$

$$|z|^2 = x^2 + y^2$$

$$u = x^2 + y^2$$

$$v = 0$$

$$\frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial v}{\partial x} = 0$$

$$\frac{\partial u}{\partial y} = 2y$$

$$\frac{\partial v}{\partial y} = 0$$

$$\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

$f(z)$ not analytic



$$f(z) = \bar{z}$$

$$z = x + iy$$

$$f(z) = x - iy$$

$$u = x$$

$$v = -y$$

$$\frac{\partial u}{\partial x} = 1$$

$$\frac{\partial v}{\partial x} = 0$$

$$\frac{\partial u}{\partial y} = 0$$

$$\frac{\partial v}{\partial y} = -1$$

$$\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

$$f(z) = (x^2 - y^2) + i 2xy \quad (= z^2)$$

$$u = x^2 - y^2$$

$$v = 2xy$$

$$\frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial v}{\partial x} = 2y$$

$$\frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial v}{\partial y} = 2x$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\therefore \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$$

continuous in D

$\therefore f(z)$ analytic in D

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 2x + i 2y$$

$$= 2(x + iy) = 2z$$

Harmonic Functions

A real-valued function ϕ

of two real variable x and y

that has continuous

first and second order partial

derivatives in domain D

satisfies Laplace's

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

equation is said to be harmonic

prove that

$$u(x, y) = x^2 - y^2 \quad \text{harmonic functions}$$

Sol

$$\frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial^2 u}{\partial x^2} = 2$$

$$\frac{\partial^2 u}{\partial y^2} = -2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 + (-2) = 0$$

$u(x, y)$ harmonic function

prove that

$$u(x,y) = \frac{y}{x^2+y^2} \text{ harmonic functions}$$

sol

$$\frac{\partial u}{\partial x} = \frac{(0)(x^2+y^2) - (2x)y}{(x^2+y^2)^2} = \frac{-2xy}{(x^2+y^2)^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{(-2y)(x^2+y^2)^2 - (2(x^2+y^2)')(2x)(-2xy)}{(x^2+y^2)^4}$$

$$= \frac{(-2y)(x^2+y^2) \left[(x^2+y^2) - 2(2x)(x) \right]}{(x^2+y^2)^4}$$

$$= \frac{-2y \left[x^2+y^2 - 4x^2 \right]}{(x^2+y^2)^3} = \frac{-2y(-3x^2+y^2)}{(x^2+y^2)^3}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{6yx^2-2y^3}{(x^2+y^2)^3}$$

$$\frac{\partial u}{\partial y} = \frac{(1)(x^2+y^2) - (2y)(y)}{(x^2+y^2)^2} = \frac{x^2+y^2-2y^2}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{(-2y)(x^2+y^2)^2 - (2(x^2+y^2)')(2y)(x^2-y^2)}{(x^2+y^2)^4}$$

$$= \frac{-2y(x^2+y^2) \left[x^2+y^2 + 2(x^2-y^2) \right]}{(x^2+y^2)^4} = \frac{-2y(3x^2-y^2)}{(x^2+y^2)^3}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{-6yx^2+2y^3}{(x^2+y^2)^3}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{6yx^2-2y^3}{(x^2+y^2)^3} + \frac{-6yx^2+2y^3}{(x^2+y^2)^3} = \frac{0+0}{(x^2+y^2)^3} = 0$$

If $f(z) = u(x, y) + i v(x, y)$

analytic in a domain D
تحليلية

then $u(x, y)$ and $v(x, y)$
are harmonic in D

prove that $u(x, y) = x^2 - y^2$
and $v(x, y) = \frac{y}{x^2 + y^2}$

are harmonic functions
توافقة

but are not harmonic conjugates

$$\frac{\partial^2 u}{\partial x^2} = 2$$

$$\frac{\partial^2 u}{\partial y^2} = -2$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\frac{\partial^2 v}{\partial x^2} = \frac{6xy - 2y^3}{(x^2 + y^2)^3}$$

$$\frac{\partial^2 v}{\partial y^2} = \frac{-6xy + 2y^3}{(x^2 + y^2)^3}$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

$$\frac{\partial u}{\partial x} = 2x$$

$$\frac{\partial v}{\partial x} = \frac{2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial v}{\partial y} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

$$\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

$f(z)$ not analytic